

OUTLOOK, SK, Canada

WMO: 715510

Lat: 51.480N

Lon: 107.050W

Elev: 1775

StdP: 13.78

Time Zone: -6.00 (W06)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.4	-17.8	-28.6	1.2	-22.2	-23.6	1.6	-17.6	26.5	22.1	24.0	24.2	7.2	280	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	87.1	65.5	83.4	64.2	80.1	62.7	69.2	80.9	67.0	78.6	65.1	75.9	10.6	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.3	100.0	74.1	63.0	92.1	72.1	61.0	85.5	70.0	34.4	80.8	32.6	78.4	31.0	75.8	78.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 23.6	20.8	18.6	DB	-29.9	93.2	5.4	3.5	-33.8	95.7	-37.0	97.8	-40.0	99.8	-44.0	102.3	(4)
(5)			WB	-30.2	73.5	5.4	2.5	-34.0	75.3	-37.2	76.8	-40.2	78.2	-44.1	80.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	39.0	10.3	12.0	24.0	40.5	52.5	60.7	66.0	64.1	55.2	41.3	26.1	13.6		
	DBStd	22.93	15.41	14.03	14.31	10.13	8.19	6.10	5.33	6.21	8.30	9.19	12.44	14.92		
	HDD50	5664	1231	1065	805	306	69	3	0	0	40	296	719	1130		
	HDD65	9682	1696	1485	1270	734	392	155	52	92	307	736	1168	1595		
	CDD50	1651	0	0	0	22	147	324	497	437	198	25	1	0		
	CDD65	193	0	0	0	0	6	26	83	64	14	0	0	0		
	CDH74	2633	0	0	0	14	164	372	937	858	280	8	0	0		
	CDH80	762	0	0	0	3	30	88	270	284	87	0	0	0		
	WSAvg	8.9	9.0	8.7	9.7	10.3	10.1	9.0	8.0	7.6	8.4	8.8	8.9	8.7		
	PrecAvg	14.3	0.6	0.4	0.6	1.1	1.8	2.9	2.3	2.0	1.3	0.9	0.5	0.5		
(15) Precipitation	PrecMax	19.7	1.5	1.0	1.5	2.6	5.2	6.7	5.0	4.4	3.9	2.4	1.6	1.2		
	PrecMin	7.4	0.0	0.0	0.0	0.2	0.2	0.7	0.3	0.3	0.0	0.2	0.0	0.0		
	PrecStd	3.1	0.3	0.2	0.4	0.7	1.3	1.4	1.3	1.0	1.0	0.5	0.4	0.3		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.6	43.1	58.3	74.8	84.0	88.4	91.7	92.2	88.3	74.2	60.2	46.9		
		MCWB	38.7	37.1	47.0	53.7	59.1	64.7	69.1	66.0	62.5	56.2	48.1	39.6		
	2%	DB	39.3	38.4	51.9	68.2	78.6	82.1	86.4	86.9	81.7	66.5	51.3	41.2		
		MCWB	35.7	34.7	43.1	50.4	57.2	62.7	67.5	65.1	60.2	51.8	42.8	36.3		
	5%	DB	36.0	35.2	46.1	62.8	74.2	78.2	82.7	82.6	76.1	60.4	45.5	37.4		
		MCWB	33.2	32.5	39.7	48.0	55.1	60.4	66.5	64.0	58.2	48.8	39.6	33.9		
	10%	DB	31.6	31.7	40.8	57.6	69.6	74.5	79.2	78.6	70.8	55.3	41.0	33.3		
		MCWB	29.6	29.8	36.5	45.7	52.9	59.0	64.9	62.6	55.8	46.2	36.9	30.7		
	0.4%	WB	39.0	37.9	48.0	55.0	62.3	69.3	73.2	69.8	64.8	57.3	48.5	40.1		
		MCDB	43.1	42.0	57.6	70.2	75.7	81.2	84.2	84.1	81.5	71.0	58.8	44.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	36.0	35.0	43.7	51.8	58.7	65.2	70.5	67.4	61.8	53.1	43.6	36.6		
		MCDB	39.1	37.8	51.0	64.7	73.7	75.8	81.6	81.1	77.5	63.8	50.5	40.8		
	5%	WB	33.3	32.7	39.9	49.1	56.5	62.8	68.3	65.7	59.3	49.8	40.0	34.0		
		MCDB	36.0	35.1	45.8	60.9	70.7	73.4	79.0	78.8	72.6	58.2	44.9	37.6		
	10%	WB	29.6	29.7	36.8	46.4	54.3	60.9	66.4	63.7	57.0	47.2	37.2	30.8		
		MCDB	31.6	31.7	40.6	56.2	67.5	71.6	76.8	75.7	69.1	54.8	40.7	33.3		
	5% DB	MDBR	16.5	16.4	16.7	20.8	23.9	20.4	21.9	23.4	22.7	19.2	15.1	14.7		
		MCDBR	18.1	18.8	22.6	31.0	31.4	27.8	28.1	31.0	32.9	28.2	21.2	18.2		
		MCWBR	15.4	16.0	14.8	16.2	14.3	11.9	12.8	13.0	14.4	15.5	14.0	14.4		
		MCWBR	18.0	17.8	21.9	28.2	27.7	22.7	24.8	26.9	29.1	25.9	20.0	17.9		
(40) Clear-Sky Solar Irradiance	taub		0.260	0.256	0.275	0.341	0.355	0.365	0.371	0.377	0.328	0.283	0.255	0.223		
		taud	2.287	2.343	2.464	2.343	2.352	2.374	2.370	2.333	2.447	2.551	2.484	2.402		
	Ebn at Noon		243	282	298	285	285	282	278	270	273	270	250	247		
		Edh at Noon	20	25	28	36	37	36	36	36	28	21	17	15		
(44) All-Sky Solar Radiation	RadAvg		360	675	1116	1514	1869	1900	2043	1643	1204	718	365	260		
	RadStd		36	48	88	136	139	141	112	116	108	98	41	25		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	-1.26	N/A	N/A	N/A	N/A	N/A	
(47) Regional (15 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page